1. What is the state file used for? Why should it not be committed to a Git repository, and how do teams usually share it instead?

   State file saves the resource data and metadata. It should not be pushed to git because it is security risks. Teams who develop the application share it through secure backend

2. Explain the difference between declarative and imperative infrastructure automation. What does idempotency mean, and why is it important?

   Declarative automation specifies the desired goal, whereas imperative specifies the step by step instructions to achieve the goal. Idempotency means running operation multiple times produces same result without side effects if target is already met.

3. What is configuration drift, how does it happen, and how does an IaC tool detect it?

   Configuration drift happens when resources actual state deviates from the code because of manual changes. Iac detects this because it compares its stored state with the actual state

4. Explain the difference between provisioning and configuration management. Where do cloud-init and Ansible fit in?

   Provisioning is used to set up the base infrastructure. Whereas configuration management installs and manages software settings. Cloud init is used in provisioning and ansible focuses on configuration management.

hands on:

Website link: http://86.50.23.135
Github link:

```
PS C:\Users\I LOVE MY GF\DOWNLOADS> cd ~
PS C:\Users\I LOVE MY GF> mkdir week7-iac-cpouta;cd week7-iac-cpouta;git init


    Directory: C:\Users\I LOVE MY GF


Mode                 LastWriteTime         Length Name
----                 -------------         ------ ----
d-----        13.10.2026     20.21                week7-iac-cpouta
Initialized empty Git repository in C:/Users/I LOVE MY GF/week7-iac-cpouta/.git/


PS C:\Users\I LOVE MY GF\week7-iac-cpouta>
```

First i create directory for week7 project and initialize empty git version control repository

```
PS C:\Users\I LOVE MY GF\week7-iac-cpouta> Get-Content .gitignore
# OpenTofu / Terraform working files and state
.terraform/
*.tfstate
*.tfstate.*
crash.log
```

Then I created .gitignore file and using Set-content command to apply there the names of the sensitive files.

```
PS C:\Users\I LOVE MY GF\week7-iac-cpouta> winget install --exact --id=OpenTofu.Tofu
The `msstore` source requires that you view the following agreements before using.
Terms of Transaction: https://aka.ms/microsoft-store-terms-of-transaction
The source requires the current machine's 2-letter geographic region to be sent to the backend service
to function properly (ex. "US").

Do you agree to all the source agreements terms?
[Y] Yes  [N] No: Y
Found OpenTofu [OpenTofu.Tofu] Version 1.13.0
This application is licensed to you by its owner.
Microsoft is not responsible for, nor does it grant any licenses to, third-party packages.
Downloading https://github.com/opentofu/opentofu/releases/download/v1.13.0/tofu_1.13.0_windows_amd64.zip
  34.3 MB / 34.3 MB
Successfully verified installer hash
Extracting archive...
Successfully extracted archive
Starting package install...
Path environment variable modified; restart your shell to use the new value.
Command line alias added: "tofu"
Successfully installed
```

Installing OpenTofu with winget. Installed OpenTofu on Windows using the command winget install --exact --id=OpenTofu.Tofu. The installation completed successfully

```
PS C:\Users\I LOVE MY GF> cd week7-iac-cpouta
PS C:\Users\I LOVE MY GF\week7-iac-cpouta> ssh-keygen -t ed25519 -f "$HOME\.ssh\week7_ed25519" -C "week7"
Generating public/private ed25519 key pair.
Enter passphrase (empty for no passphrase):
Enter same passphrase again:
Your identification has been saved in C:\Users\I LOVE MY GF\.ssh\week7_ed25519
Your public key has been saved in C:\Users\I LOVE MY GF\.ssh\week7_ed25519.pub
The key fingerprint is:
```

Then i generated a new ED25519 SSH key pair for accessing the virtual machine using the ssh-keygen command. The private and public keys were saved to the $HOME\.ssh\

```
# Look up your existing project network (created by CSC, not by us)
data "openstack_networking_network_v2" "project" {
  name = var.project_network
}

# SSH key pair: only the PUBLIC key is uploaded
resource "openstack_compute_keypair_v2" "web" {
  name       = "${var.name_prefix}-key"
  public_key = file(pathexpand(var.ssh_public_key_path))
}

# Firewall (security group) with only the rules we need
resource "openstack_networking_secgroup_v2" "web" {
  name        = "${var.name_prefix}-web-sg"
  description = "SSH from my IP, HTTP from anywhere. Managed by OpenTofu."
}

resource "openstack_networking_secgroup_rule_v2" "ssh" {
  security_group_id = openstack_networking_secgroup_v2.web.id
  direction         = "ingress"
  ethertype         = "IPv4"
  protocol          = "tcp"
  port_range_min    = 22
  port_range_max    = 22
  remote_ip_prefix  = var.ssh_allowed_cidr
}

resource "openstack_networking_secgroup_rule_v2" "http" {
  security_group_id = openstack_networking_secgroup_v2.web.id
  direction         = "ingress"
  ethertype         = "IPv4"
  protocol          = "tcp"
  port_range_min    = 80
  port_range_max    = 80
  remote_ip_prefix  = "0.0.0.0/0"
}

# Network port for the VM in the project network, protected by the security group
resource "openstack_networking_port_v2" "web" {
  name               = "${var.name_prefix}-port"
  network_id         = data.openstack_networking_network_v2.project.id
  admin_state_up     = true
  security_group_ids = [openstack_networking_secgroup_v2.web.id]
}

# The virtual machine. cloud-init (user_data) installs Apache at first boot.
resource "openstack_compute_instance_v2" "web" {
  name        = "${var.name_prefix}-vm"
  image_name  = var.image_name
  flavor_name = var.flavor_name
  key_pair    = openstack_compute_keypair_v2.web.name

  # Fill the template and normalise Windows line endings (CRLF -> LF)
  user_data = replace(templatefile("${path.module}/cloud-init.yaml", {
    student_name = var.student_name
    page_message = var.page_message
  }), "\r\n", "\n")

  network {
    port = openstack_networking_port_v2.web.id
  }
}

# Public (floating) IP from the "public" pool, attached to the VM's port
resource "openstack_networking_floatingip_v2" "web" {
  pool    = "public"
  port_id = openstack_networking_port_v2.web.id
}
```

Why architecture is made this way:
Resources refer to each other using attributes like .id and .name (both the virtual machine and the floating IP refer to the network port ID). This allows OpenTofu to automatically map dependencies and create resources in exactly the right order.

Here is the code blocks above explained:

main.tf Structure: Data Source (data): Retrieves an existing project network provided by cPouta.

Key Pair (openstack_compute_keypair_v2): Exports local public SSH key to the cloud for secure connection.

Firewall Group (openstack_networking_secgroup_v2): Creates the firewall to restrict SSH connection to your specified IP address only and leaves HTTP traffic open to everyone.

Network Port (openstack_networking_port_v2): Creates a virtual network port connected to the project network and applies firewall rules to it.

Virtual Machine (openstack_compute_instance_v2): Creates an Ubuntu virtual machine and runs the cloud-init script to automatically install a web server.

Floating IP (openstack_networking_floatingip_v2): Allocates a public IP address and binds it to the network port of the virtual machine.

```
PS C:\Users\I LOVE MY GF\week7-iac-cpouta> Invoke-RestMethod https://ifconfig.me/ip
```

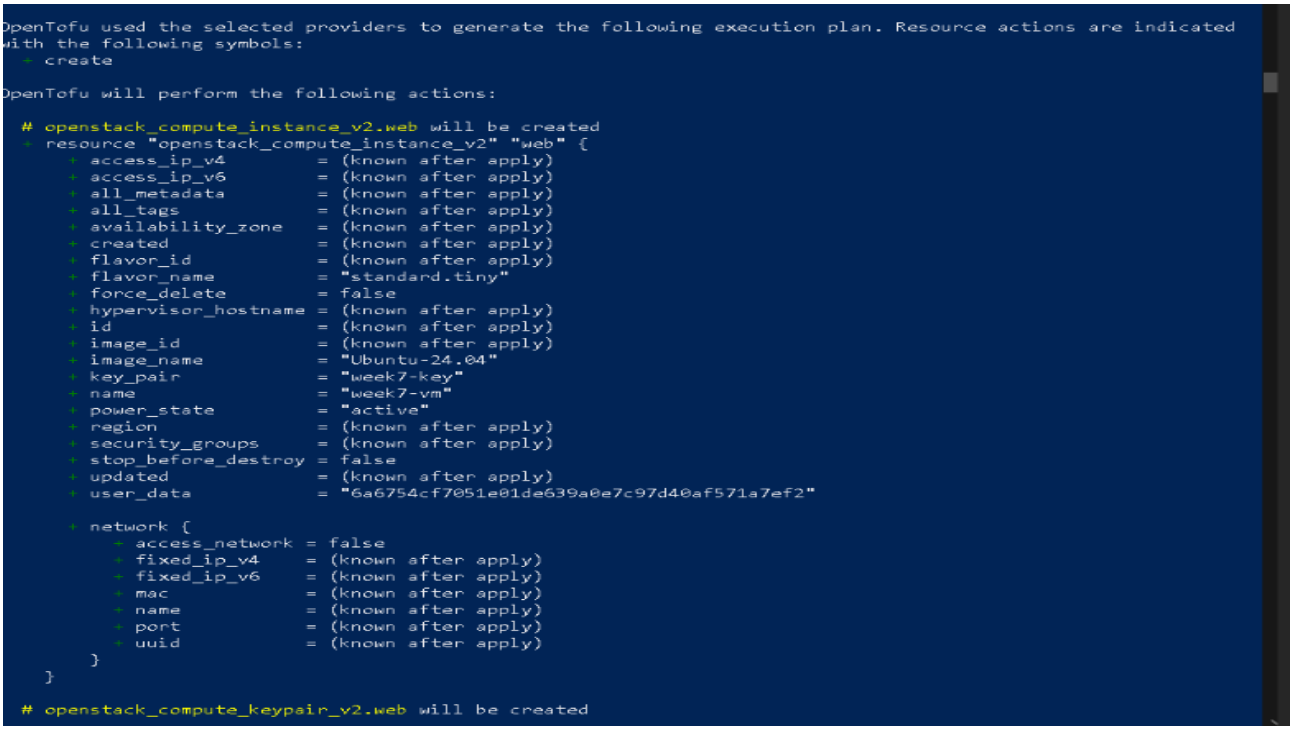
I check my own ip address with this. Checking the local ip address is necessary to restrict SSH access in the security group rules. So we ensure that only my personal workstation can connect to virtual machine via ssh.

```
PS C:\Users\I LOVE MY GF\week7-iac-cpouta> Set-Content -Path terraform.tfvars -Value @"
>> project_network      = "project_2020487"
>> student_name         = "Santeri Kreivi"
>> page_message         = "Week7 works!"
>> "@
```

I created and filled the terraform.tfvars file using PowerShell to set project-specific variables, including my student name and custom web page message. This keeps the OpenTofu code reusable and prevents personal and sensitive data from being committed to Github.

```
Length Name
------ ----
   240 .gitignore
   661 cloud-init.yaml
   848 clouds.yaml
  1829 main.tf
   286 outputs.tf
   163 terraform.tfvars
   984 variables.tf
   227 versions.tf
```

Then I verified the project directory structure using get-ChildItem in PowerShell and made sure that all of the required files are there. Filling the configuration files required a lot of effort but AI assistance and reading tutorials helped me to pull it through.



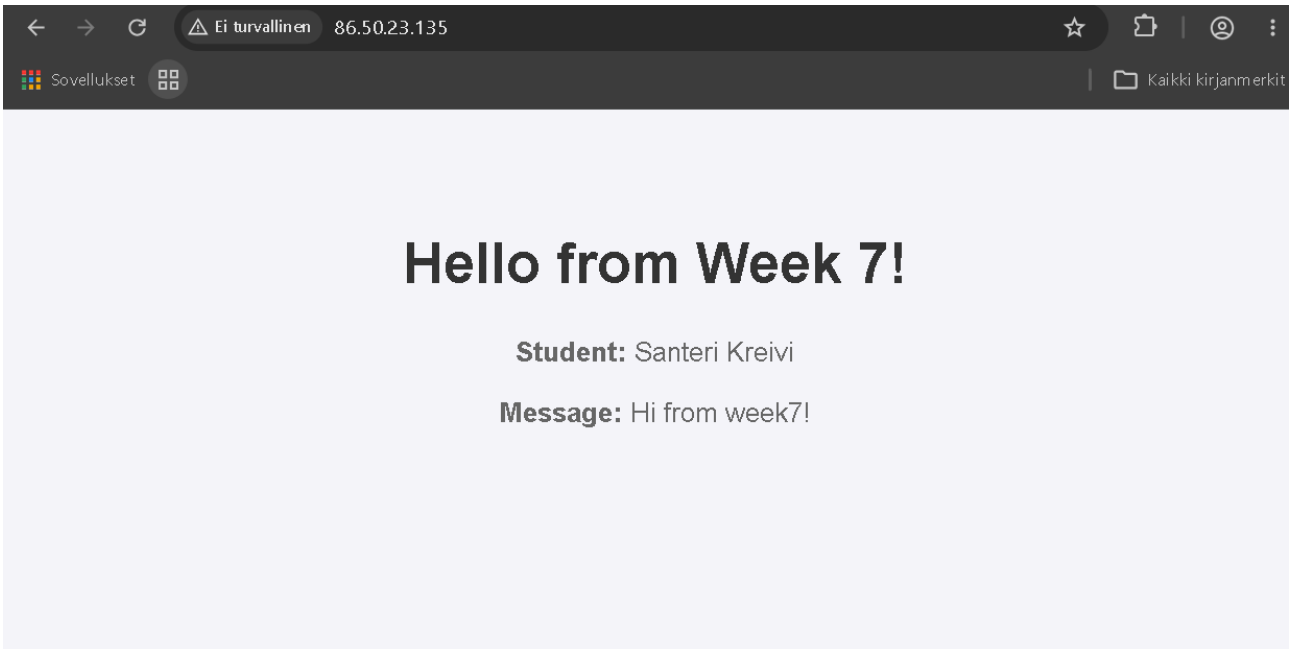
I ran tofu plan to check execution plan before creating the resources. The output verifies that opentofu is set to provision the Ubuntu 24.04 virtual machine (week7-vm) along with the required key pairs and network settings.

```
Apply complete! Resources: 7 added, 0 changed, 0 destroyed.

Outputs:

floating_ip = "86.50.23.135"
web_url = "http://86.50.23.135"
PS C:\Users\I LOVE MY GF\week7-iac-cpouta>
```

A succesful tofu apply!



This verified the succesful deployment. This is the page generated by cloud-init correctly rendering my name and custom message.